Extraction and Access to Information in Natural Language for Non-Developers - Democratizing Information

Progress Report – 2023/2024

Emanuel Matos

DOCTORAL PROGRAMME IN COMPUTER SCIENCE

University of Aveiro
Portugal
September - 2024

Contents

Chapter 1

Introduction

Extracting information automatically from natural language sources has a variety of applications, such as Business Intelligence [1], Forensics [2], Medicine [3], and question answering systems [4]. Named Entity Recognition (NER) is crucial for information extraction. Developing NER for new domains heavily relies on annotated data; thus, our work aims to contribute methods for enabling NER in domains that lack such datasets.

Somewhat constrained by being a part-time student based in Brazil, in 2023-2024, we concentrated on diversifying our methods across different environments and datasets and exploring new ideas. We delved into the realm of emerging methods for NER, namely generative approaches based in Large Language Models (LLMs), assessed numerous options, and achieved technical milestones, which allowed us to publish and present in international conference(s). Through these experiments, we developed a new pipeline that will underpin our research next year.

This report outlines the efforts completed during the fourth year of our thesis work, emphasizing our advancements in enhancing NER capabilities in new domains. As a part-time student, this 4rd year corresponds to the first semester of the 3rd year of a full time student.

This short progress report is structured as follows: next chapter presents main developments; chapter 3 the main results; chapter 4 the revised workplan.

Chapter 2

Main Developments

Last year the work had good progress, resulting of 5 main lines of work:

Automatic creation of NER Systems for new domains – An innovative approach for developing Named Entity Recognition (NER) systems in new domains without the need for annotated data was conceive, implemented and evaluated.

The method, profiting from previous years developments, uses automatically created annotations to fine-tune deep learning models like BERT, RoBERTa, and BART, achieving over 75% accuracy across various entity sets.

Using an Open Information Extraction system and an automatic annotation mapping mechanism, we showcased the method's efficacy utilizing the WikiNER dataset.

Generative NER – In the second part of the year, motivated by recent developments in NER (and many other NLP tasks), we initiated exploration of Generative approaches and In-context Learning to NER.

Initial exploration include acquisition of knowledge in relevant topics such as Large Language Models, In-context Learning and prompt engineering, as well as revision of related literature.

Next step consisted in exploratory experiments and search for tools to support systematic research with minimal costs. The initial experiments were conducted using a recently released large language model (LLM) specifically developed for Brazilian Portuguese, known as Maritalk's Sabiá-2.

Sabiá-2 is a suite of advanced language models tailored to Portuguese, with a particular emphasis on Brazilian cultural and linguistic contexts. Extensive research and simulations to investigate new learning models aiming for state-of-the-art performance in emerging technologies were performed. Our primary objective was to enhance the performance of named entity recognition (NER) systems by using advanced techniques and algorithms.

Based on the preliminary experimental results obtained, very promising, we focused in the development and evaluation of a complete NER system based on In-context learning without resorting to manually annotated data.

We instantiated the system with two distinct large language models. We employed Google's Bison Maritalk's Sabia-2 to accurately interpret and process our dataset sentences. Bison's robust architecture notably improved the accuracy of our NER systems. Specifically designed for the Portuguese language, Sabia-2 also increased the efficiency and robustness of our NER systems by providing specialized understanding and processing of linguistic patterns unique to this language.

Our strategy involved the utilization of two types of prompts: basic prompts and XLM. Our objective was to analyze different prompt types in terms of their effectiveness in identifying and classifying entities in Portuguese texts. Two main types were investigated:

Type 1 - "Classic" Prompts: These prompts present the task in a straightforward question-and-answer style. They are designed to enhance clarity and simplicity, allowing the model to identify entities by directly asking about the type of entity within a sentence. The objective of Type 1 prompts is to capture a broader range of entities, thereby improving recall.

Type 2 - XML-based: This category of prompts uses a more structured format by embedding entities within XML tags. This structured approach helps the model accurately determine the exact boundaries of entities, thereby increasing precision. XML tags clarify where an entity begins and ends, minimizing potential ambiguity in the model's predictions.

By merging these prompts and algorithms, we achieved significant enhancements in the accuracy and efficiency of our NER systems, harnessing the strengths of both basic and sophisticated language processing techniques.

Expansion of Datasets used – This last year we continued to work with our initial datasets and introduced a new language (WikiNER Spanish) , with the aim of creating language diversity. This is necessary to answer questions about the theory/process we are developing.

The WikiNER [5], created by Nothman, contains 7.200 manually labeled Wikipedia articles in nine multilingual languages: English, German, French, Polish, Italian, Spanish, Dutch, Portuguese, and Russian.

Assessment of Cross-language potential of Generative NER – A Spanish dataset, WikiNER SP, was used to test the cross-language capability of models trained on Portuguese data. This dataset provides a comprehensive test environment for assessing the models' ability to adapt across languages, underlining the challenges and success of cross-language adaptation methods.

Publication of results - Our research results were documented and submitted to international conferences, aiming to disseminate our findings to a wider audience. Based on these findings, we prepared and submitted a paper to IberSpeech 2024, consolidating the insights gained over the year. This submission presents our work to a prestigious conference, contributing to the ongoing discourse in speech and language processing.

More information regarding this subject is presented in next chapter, on results.

Chapter 3

Results

This chapter highlights the main results obtained last year. They integrate publications, presentations and software developments.

3.1 Main Results

Publication and Presentation – During the period one submission to international conference was accepted, and presented.

The paper "Towards the Automatic Creation of NER Systems for New Domains" [6] was submitted to **PROPOR 2024**, the main conference in NLP for Portuguese language, an accepted as a full paper. It addresses the significant challenge of developing Named Entity Recognition (NER) systems for new domains that lack annotated data. The proposed approach leverages Open Information Extraction (OpenIE) and fine-tuning of deep learning models to enhance the adaptability of NER systems using the BERT model with automatically generated annotations. This innovative method employs Open IE alongside a mapping mechanism to provide automatic annotations for fine-tuning, yielding promising results with accuracy rates around 75% for entity sets comprising 20 elements. The paper examines various factors influencing performance, including the choice of deep learning models, the number of entities, and the size of the training data set.

The paper was presented in one of the oral sessions by the author, with very positive comments from the audience.

The paper was published in ACL Anthology (`https://aclanthology.org/2024.propor-1.22/`)

Submissions – During the period one additional submission to international conference was made. Notification of results is planned for the 20th of september.

The paper entitled "OPENER - Open-NER for Domains Lacking Annotated Resources", resulting from the Generative NER research line, was submitted o IberSPEECH 2024 (11 to 13 of November 2024). The paper submitted introduces a novel technique for creating Named Entity Recognition (NER) systems in domains that lack annotated datasets, employing Large Language Models (LLMs) and automatic annotation methods. By utilizing in-context learning with few-shot samples from pre-existing NER systems, the method eliminates the need for manual annotations. The research assesses this technique using two datasets (WikiNER and MiniHAREM) and two LLMs (Bison and Sabiá-2) in both Portuguese and Spanish. The results reveal high median Precision scores exceeding 75% and an F1 score reaching 78.3% for WikiNER PT with Bison. Nevertheless, cross-domain experiments indicate a reduction in Recall to below 20%. The insights suggest that augmenting the number of examples in prompts enhances Precision, Bison surpasses Sabiá-2, and both LLMs are capable of processing Spanish with Portuguese examples. The method demonstrates flexibility across languages and domains, indicating a quicker adaptation to new domains without the necessity for manually annotated data.

New NER systems for Portuguese – To the Transformer-based system that was evolved in part of 2023-2024, a new NER pipeline was created, that we named OPENER, Open-NER for domains lacking annotation resource.

Draft of parts of the Thesis – Mainly by integration and adaptation of published and submitted papers and pre-thesis and progress reports, the basis of the future Thesis is also a relevant result of 2023-2024. At time of writing it has more than 60 pages.

3.2 Comments

The publications planned for the year 2023-2024 were achieved.

Advances were made in models and algorithms available for experiments, processing pipeline and new ways to evaluated automatic annotations of NER systems.

Work has deviated from the plan regarding the initial idea of making a initial system with basic modules that would be later improved. Work is now focused on NER and a more bottom-up approach, developing modules with good individual performance. This new focus will probably imply a request for alteration of the Thesis title in the future.

Chapter 4

Future work

Taking into consideration our results and previous work plan, the year 2023-2024 was in line with our main focus: to study and create the building blocks for automating the capture and tagging of entities (NER).

We move towards our objective, the next steps considered essential for the development of the work are:

- **Refactor/improve the implementations of our 2 types of NER systems to enable more extensive assessment.**
- **Evaluation of the OPENER with new domains (including cross-language if possible).**
- **State-of-the-art update.**
- **Integration in the Thesis of submitted paper.**
- **Preparation of a journal paper.**
- **Conclusion of Thesis writing.**
- **Thesis submission.**

4.1 Updated Workplan

Drawing from our experiences in 2023/2024, where part-time work and frequent daily interruptions posed challenges, we are confident that the adjustments we propose to our plan will result in more consistent work and meaningful academic contributions. Our accumulated experience, coupled with the invaluable support of our supervisors, has created a well-balanced foundation, making these adjustments a logical step for the upcoming periods. The reorganized plan is detailed in the Table 4.1.

Task	2024-2025			
	1Q	2Q	3Q	4Q
State-of-the-art update	X	X	X	
Full Script[1]	X	X	X	
NER 4 new domains [2]	X	X		
Custom class sets in NER[3]	X	X		
Publications	X	X		X
Thesis	X	X	X	X

Table 4.1: Revised Workplan.

4.1.1 Milestones - Update

According to the changes made to the workplan, Milestones were also updated, as follows[4]:

- M3 - NER for custom class sets and new domains – Resulting directly from the revised workplan is work in progress with conclusion planned for no later than the 2nd Quarter of 2024-2025.
- M4 - Document detailing the work carried out and the scientific results obtained to support the proposed research questions. Planned for September 2025.

[1]RQ: OpenNer can provide a NER useful to create automatic annotations?
[2]RQ: our approach generalizes to other domains?
[3]RQ: Can we go beyond PERSON, LOC, ORG classic class sets with automatically annotated data?
[4]M1 and M2 of our initial worplan were already achieved.

Chapter 5

Conclusion

We consider that research is advancing in the correct trajectory, opinion substantiated by the feedback garnered from the paper presentation at PROPOR 2024 in Santiago de Compostela.

Based on the analysis of the results and the evolution of our the research, we propose minor adjustments to workplan, without compromising its foundational principles.

Aligning with our overarching objectives, the roadmap for 2024-2025 remains unchanged from the one established last year. Our focus will be directed towards the development of core modules and the creation of Named Entity Recognition (NER) systems for new domains.

Bibliography

[1] K. Sintoris and K. Vergidis. Extracting business process models using natural language processing (NLP) techniques. In *2017 IEEE 19th Conference on Business Informatics (CBI)*, volume 01, pages 135–139, 2017.

[2] Flora Amato, Giovanni Cozzolino, Vincenzo Moscato, and Francesco Moscato. Analyse digital forensic evidences through a semantic-based methodology and NLP techniques. *Future Generation Computer Systems*, 98:297–307, 2019.

[3] Antonio Moreno Sandoval, Julia Díaz, Leonardo Campillos Llanos, and Teófilo Redondo. Biomedical Term Extraction: NLP Techniques in Computational Medicine. *IJIMAI*, 5(4):51–59, 2019.

[4] Hiral Desai, Mohammed Firdos Alam Sheikh, and Satyendra K Sharma. Multi-purposed question answer generator with natural language processing. In *Emerging Trends in Expert Applications and Security*, pages 139–145. Springer, 2019.

[5] Joel Nothman, Nicky Ringland, Will Radford, Tara Murphy, and James R Curran. Learning multilingual named entity recognition from Wikipedia. *Artificial Intelligence*, 194:151–175, 2013.

[6] Emanuel Matos, Mário Rodrigues, and António Teixeira. Towards the automatic creation of NER systems for new domains. In *Proceedings of the 16th International Conference on Computational Processing of Portuguese*, pages 218–227, 2024. `https://aclanthology.org/2024.propor-1.22/`.